

Exploiting Opponents in Drop the Handkerchief

James Cenawood

September 29, 2026

In the registered v1 study, translated Hal wins **93.9%** against player emulators, where exact Hal wins 54.7%. A guard caps Hal’s expected loss against **any** opponent, but no guarded Hal beat translated Hal in a registered test.

1. From solving to exploiting

Exact Hal plays the certified equilibrium of Drop the Handkerchief (DTH) [1]. No opponent does better than the game value against it, but it gains nothing from a player’s habits. We build Hal policies that forecast the opponent from the public history and answer that forecast in the exact stage game.

At state x , the tablebase supplies the continuation-adjusted matrix $M(x)$ and the value $W(x)$ [1, Section 4]. In Hal’s view, let $H = M$ when Hal drops and $H = -M^T$ when Hal checks, so rows are Hal’s seconds and entries are Hal’s values; W_H is the value in the same view. Given a forecast q of the opponent’s second, Hal plays a hard best response to Hq , with equal mass on tied seconds. The policies below differ in how they form q and in the matrix they answer.

2. Perfect Mode Hal

Perfect Mode (PM) Hal fuses forecast sources for each opponent role: the uniform policy, the opponent’s equilibrium, a decayed Dirichlet count, Perfect Hal v1’s expert mixture, change-point detection [5], an outcome model, and an optional recurrent network. Fixed Share [4] learns their weights. Four modes move from the equilibrium to hard forecast responses, and we measure each chosen policy’s local worst-case loss against a fresh matrix.

We registered a 56-identity confirmation in commit b76147f. PM scored .8705, behind Perfect Hal v1 at .9554, and failed its promotion gate (Table 1). Its record `pm_hal_confirmation_v3.json` sits in `src/arena/config`. Later builds replaced the tablebase manifest that it binds, so we do not rerun v3. PM plays local pure-DTH games; the hosted game does not offer it.

3. Translated Hal

Translated Hal adds eight offset experts to Perfect Hal v1 (Old). Each copies or mirrors one of Hal’s recent seconds with a learned offset. It beat Old by 2.11 points [1.55, 2.66] over 8,192 synthetic games and tied Old against emulators of 8 recorded players. A selector network that reweights its 24 experts added 1.52 points [0.49, 2.59] (Table 2). We did not register either study in Git before it ran, so we treat both as development evidence.

4. Exploit continuation

M assumes equilibrium play after the current half-round. Hal keeps exploiting, so M undervalues the states where Hal survives. We replace the value $W(y)$ of each live child y by the value of Hal’s best response to a child forecast q' , in the child Dropper’s view:

$$v(y) = \begin{cases} \max_d (M(y) q')_d, & \text{Hal drops at } y, \\ \min_c (q'^T M(y))_c, & \text{Hal checks at } y. \end{cases}$$

We rebuild the look-ahead matrix L from these values with the transition classes of [1, Section 4]. At each child, a best response earns at least the maximin value, which lies within 10^{-6} of $W(y)$. Thus $L \geq H$ entry by entry in Hal’s view, up to that tolerance. The *same* forecast reuses q at each child; the *role* forecast asks the model for the role the opponent takes there. A decision rebuilds up to 61 child matrices in about 25 ms.

Table 1: PM v3 confirmation, 112 games per controller.

Controller	Score	W–L–S
PM Hal	.8705	97–14–1
Perfect Hal v1	.9554	107–5–0
PM without Aggro	.8259	92–19–1
Aggro Hal	.7232	81–31–0
Adaptive Hal	.7188	80–31–1
Exact Hal	.4955	54–55–3

Table 2: Development evidence: new minus base score, with 95% identity-clustered intervals.

New vs. base	Games	Difference
Translated vs. Old, league	8,192	+ .021 [.016, .027]
Translated vs. Old, emulators	5,120	+ .001 [−.003, .006]
Selector vs. translated	2,432	+ .015 [.005, .026]

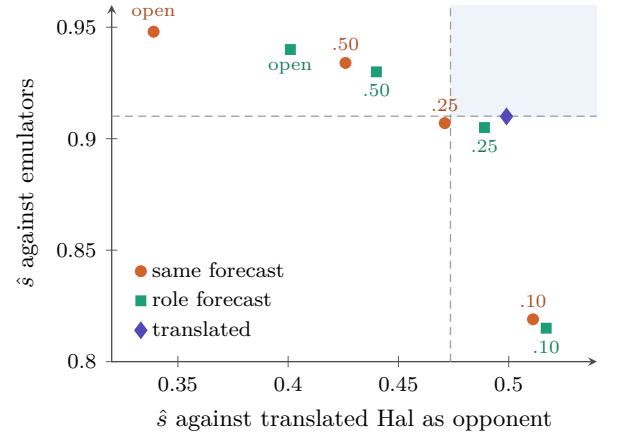


Figure 1: Selection by arm and guard ϵ_0 . The shaded corner holds arms that beat translated Hal on the emulators and stay within .025 of it against the Hal model. No arm reached it.

5. A guard on realized gifts

L trusts q' at every child, so an opponent who predicts Hal can punish it. Following Ganzfried and Sandholm [3], a guard gives each game a budget $k_0 = \epsilon_0$. Hal plays its best response to Lq when $\min_b(\pi^\top H)_b \geq W_H - \max(k, 0)$; otherwise a linear program finds the best policy that meets this floor, with the equilibrium as a fallback. After the opponent reveals b , the budget absorbs the gift

$$g = \pi^\top H e_b - W_H, \quad k \leftarrow k + g.$$

Bound. In pure DTH, a guarded Hal has $\mathbb{E}[G] \geq W_H(s_0) - \epsilon_0 - \mathbb{E}[S]$ against any opponent, where $G = \pm 1$ is the outcome and S sums the turns' floor misses.

Proof. Each transition raises the progress score Φ [1, Section 3], so play ends at a terminal state with $W_H = G$, and

$$G - W_H(s_0) = \sum_t (W_H(s_{t+1}) - W_H(s_t)).$$

The opponent commits b_t without seeing Hal's second, and Hal draws its second from π_t , so term t has conditional mean g_t . The floor gives $g_t \geq -\max(k_t, 0) - m_t$, so $k_{t+1} \geq \min(k_t, 0) - m_t$ and $\sum_t g_t = k_T - \epsilon_0 \geq -\epsilon_0 - S$ on every path. QED

The certificate can place the maximin of H up to 10^{-6} below W_H , so a turn can miss the floor by m_t . Confirmation measured misses of at most 9.8×10^{-10} . Leap turns use the equilibrium and leave k unchanged, so the bound covers pure DTH.

6. A registered test

We score a finished pure-DTH game by

$$\hat{s} = \frac{1}{2} \left(1 + W_H(s_0) + \sum_t (\pi_t^\top H_t \mu_t - W_H(s_t)) \right),$$

where μ_t is the simulated opponent's mixed policy. The same telescoping makes $\hat{s}$ an unbiased estimate of the score. We average one-step action and revival outcomes; sampled paths contribute variance. We pushed the runner with a frozen protocol in commit `b8f87b4` before any game. Sessions play four games from 8:12 in paired seats against four opponent blocks: 19 league families, emulators of 8 recorded players, translated Hal as the opponent, and a counter. The counter copies the controller from the public history and answers its exact mixed policy. Selection kept the arms that match translated Hal on the league and stay within .025 of it against the Hal model, then picked the best on the emulators. Confirmation used fresh seeds, and promotion needed every gate in Table 3.

7. Results

In canonical STL [2] against the emulators, translated Hal scored .939 (11,543–745) and exact Hal .547 (6,718–5,570). The player-clustered difference is +.393 [.370, .407].

The open arms gained 3 to 4 points on the emulators but fell to .34–.40 against the Hal model (Figure 1). Guards held the Hal model near .5, and against the counter each guarded arm reached its bound $.5 - \epsilon_0/2$. One arm passed selection: the role forecast with $\epsilon_0 = .25$. In confirmation it gained 3.5 points on the league and tied translated Hal on the emulators, but it failed three gates. We therefore deploy translated Hal.

Table 3: Confirmation of the role forecast with $\epsilon_0 = .25$ minus translated Hal. A gate passes when the 95% lower bound exceeds the limit.

Gate	Block	Mean	Lower	Limit	
P1	Emulators, $\hat{s}$	+.0001	-.017	0	fail
P2	League, $\hat{s}$	+.035	+.026	-.005	pass
P3	Hal model, $\hat{s}$	-.004	-.029	-.025	fail
P5	Canonical, raw	-.003	-.022	-.010	fail

The bound held (P4): over 6,976 games the least gift sum was $-.2500000001$. The hosted runtime check passed (P6) with a decision p95 of 24.9 ms. Against its counter, translated Hal lost all 512 confirmation games. Its public code lets a player compute Hal's mixed policies. The emulators fit 98 games against exact Hal and contain no reaction to exploitation. The v1 self-repeat fit used observed repeat rates as copy weights, which increased repetition. We retain these historical results; the corrected v2 fit needs a new study. A human win-rate claim needs a live randomized comparison.

References. [1] J. Cenawood, [Solving Drop the Handkerchief](#), September 11, 2026. [2] J. Cenawood, [Solving Surpassing the Leader](#), September 21, 2026. [3] S. Ganzfried and T. Sandholm, [Safe Opponent Exploitation](#), ACM TEAC, 2015. [4] M. Herbster and M. Warmuth, [Tracking the Best Expert](#), Machine Learning, 1998. [5] R. Adams and D. MacKay, [Bayesian Online Changepoint Detection](#), 2007.